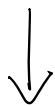


LOT-SIZING STOCASTICO



NEWSVENDOR MODEL

Hp.

- single-period → $\bar{X}$ = domanda (var. aleatoria)
- no backorder → lost sales
- single-item

ordinare troppo
↓
OVERAGE COST

ordinare troppo poco
↓
SHORTAGE COST

Notazione

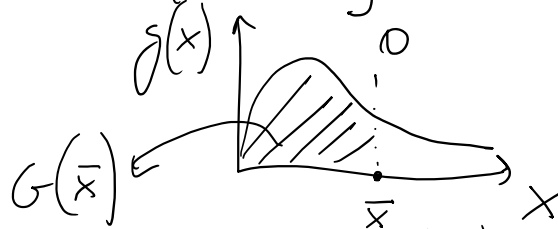
$$g(\bar{x}) = \frac{d}{d\bar{x}} = \frac{d G(\bar{x})}{d\bar{x}}$$

$$G(\bar{x}) = \int_0^{\bar{x}} g(x) dx$$

c_o = costo unitario overage

c_s = costo unitario shortage

Q = quantità da ordinare (var. decisionale)



$Y(Q) =$ funzione di costo

$$= C_0 \cdot E[N_0] + C_s \cdot E[N_s]$$

$$N_0 = \begin{cases} Q - X & \text{se } Q \geq X \\ 0 & \text{se } Q < X \end{cases}$$

$$N_s = \begin{cases} X - Q & \text{se } X \geq Q \\ 0 & \text{se } X < Q \end{cases}$$

$$N_0 = \max \{Q - X, 0\}$$

$$N_s = \max \{X - Q, 0\}$$

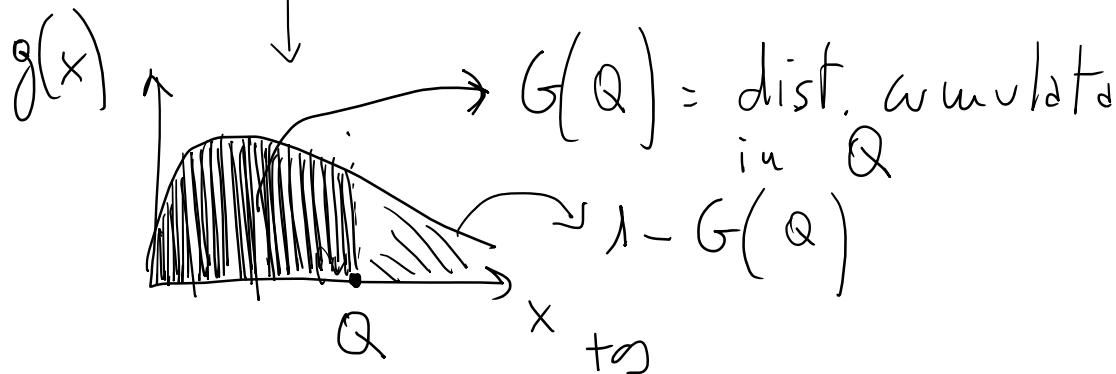
$$Y(Q) = C_0 \cdot \int_0^{+\infty} \max \{Q - x, 0\} \cdot g(x) dx + C_s \cdot \int_0^{+\infty} \max \{x - Q, 0\} \cdot g(x) dx$$

Spezzo gli integrali tra $[0, Q]$ e $[Q, +\infty]$

$$Y(Q) = C_0 \cdot \left[\int_0^Q \max \{Q - x, 0\} \cdot g(x) dx + \int_Q^{+\infty} \max \{Q - x, 0\} \cdot g(x) dx \right]$$

$$\begin{aligned}
 & + C_5 \cdot \left[\int_0^Q \max\{x-Q, 0\} g(x) dx + \int_Q^{+\infty} \max\{x-Q, 0\} \cdot g(x) dx \right] \\
 & = C_0 \cdot \int_0^Q (Q-x) \cdot g(x) dx + C_5 \int_Q^{+\infty} (x-Q) \cdot g(x) dx
 \end{aligned}$$

$$\frac{\partial y(Q)}{\partial Q} = c_o \cdot \int_0^Q g(x) dx + c_s \cdot \int_Q^{+\infty} -g(x) dx$$



$$= c_o \cdot G(Q) - c_s \cdot \int_Q^{+\infty} g(x) dx$$

$$= c_o \cdot G(Q) - c_s \cdot [1 - G(Q)] = 0$$

$$\Rightarrow \boxed{G(Q^*) = \frac{c_s}{c_s + c_o}}$$

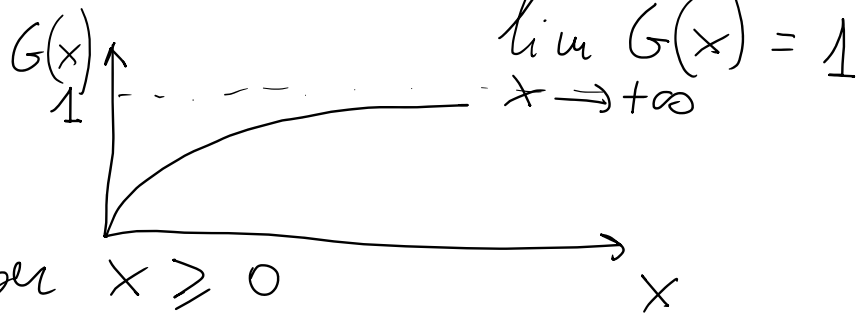
$$G(Q^*) = \Pr \{X \leq Q^*\} = \frac{c_s}{c_s + c_o}$$

Es. $X \sim \exp$

$$g(x) = \lambda e^{-\lambda x}$$

con $\lambda = \text{rate}$

$$G(x) = 1 - e^{-\lambda x}$$



$$g(x) = \begin{cases} \lambda e^{-\lambda x} & \text{per } x \geq 0 \\ 0 & \text{per } x < 0 \end{cases}$$

$$G(Q^*) = 1 - e^{-\lambda Q^*} = \frac{c_s}{c_s + c_o} = \alpha$$

$$\Rightarrow Q^* = \frac{-\ln(1 - \alpha)}{\lambda} \quad Q^* = f(c_s, c_o, \lambda)$$

Es.

• Domande t-shirt $\sim \exp^{-}(\lambda)$

Proprietà $\exp^{-}$: $\lambda = \frac{1}{\overline{X}}$

• $\overline{X} = 1000 \Rightarrow \lambda = \frac{1}{1000}$

• Costo acquisto = 10 €
 • Costo vendita = 15 €
 • Costo "rendita" = 8 €

$$\Rightarrow \begin{cases} C_0 = 10 - 8 = 2 € \\ C_S = 15 - 10 = 5 € \end{cases}$$

$$d = \frac{C_S}{C_0 + C_S} = \frac{5}{7} = 0.714$$

$$G(Q^*) = 1 - e^{-\frac{Q^*}{1000}} = 0.714 \Rightarrow Q^* = 1253$$

Se le t-shirt restano invendute:

$$\begin{cases} C_0 = 10 \\ C_S = 5 \end{cases} \Rightarrow G(Q^*) = 1 - e^{-\frac{Q^*}{1000}} = \frac{5}{15} = 0.33 \Rightarrow Q^* = 405$$